

AI Disease Risk Prediction System

Viva Preparation Notes

Q1. What is your project?

Our project is an AI Disease Risk Prediction System that predicts whether a person is at risk of a disease based on health parameters using Machine Learning and a Streamlit dashboard.

Q2. Why did you choose this project?

Healthcare is an important field where early disease prediction can help people take preventive measures. We also wanted to learn Machine Learning through a real-world problem.

Q3. What problem does your project solve?

It provides quick preliminary disease risk prediction using health data.

Q4. Is your project AI or Machine Learning?

It is primarily a Machine Learning project. Machine Learning is a subset of AI.

Q5. Which dataset did you use?

Pima Indians Diabetes Dataset and Heart Disease Dataset.

Q6. Where did you get the dataset?

Kaggle/UCI Machine Learning Repository.

Q7. How many records were present?

The Diabetes dataset contains 768 records.

Q8. What is the target variable?

Outcome: 0 = No Disease, 1 = Disease.

Q9. What are features?

Input variables such as Glucose, BMI, Age, Blood Pressure, etc.

Q10. What is Data Preprocessing?

Cleaning and preparing raw data before model training.

Q11. Why is preprocessing necessary?

To improve data quality and model performance.

Q12. Did you have missing values?

We checked and handled data quality before training.

Q13. What is data cleaning?

Removing errors, duplicates, and inconsistencies.

Q14. What is EDA?

Exploratory Data Analysis used to understand patterns and relationships in data.

Q15. Which visualizations did you use?

Count Plot, Histogram, Correlation Heatmap.

Q16. Why use a Heatmap?

To identify relationships between features.

Q17. What is Machine Learning?

A branch of AI that learns patterns from data and makes predictions.

Q18. Which type of Machine Learning did you use?

Supervised Learning.

Q19. Why supervised learning?

Because the dataset contains labeled outputs.

Q20. Which algorithms did you use?

Logistic Regression and Random Forest.

Q21. Why Logistic Regression?

It works well for binary classification problems.

Q22. Why not Linear Regression?

Linear Regression predicts continuous values, while our output is categorical.

Q23. Why Random Forest?

It combines multiple decision trees and reduces overfitting.

Q24. Which model performed better?

Answer with your final project result and accuracy.

Q25. Why split the dataset?

To evaluate model performance on unseen data.

Q26. What split ratio did you use?

80% training and 20% testing.

Q27. What is training data?

Data used to train the model.

Q28. What is testing data?

Data used to evaluate the model.

Q29. What accuracy did your model achieve?

Approximately 79% (use exact final value in viva).

Q30. What is accuracy?

Percentage of correctly predicted observations.

Q31. What is a Confusion Matrix?

A table comparing actual and predicted values.

Q32. What are TP, TN, FP, FN?

True Positive, True Negative, False Positive, False Negative.

Q33. What is Precision?

Out of predicted positives, how many were actually positive.

Q34. What is Recall?

Out of actual positives, how many were correctly identified.

Q35. What is F1 Score?

The harmonic mean of Precision and Recall.

Q36. What is Streamlit?

A Python framework for building interactive ML web apps.

Q37. Why use Streamlit?

It is simple, fast, and requires minimal frontend coding.

Q38. Could this be deployed online?

Yes, using Streamlit Cloud, Render, Railway, AWS, etc.

Q39. Which libraries did you use?

Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Joblib, Streamlit.

Q40. What is Pandas?

A Python library for data manipulation and analysis.

Q41. What is NumPy?

A Python library for numerical computing.

Q42. Why use Joblib?

To save and load trained ML models.

Q43. What is Overfitting?

When a model memorizes training data and performs poorly on new data.

Q44. What is Underfitting?

When a model fails to learn important patterns.

Q45. If accuracy is 100%, is that always good?

No, it may indicate overfitting.

Q46. How would you improve the project?

More datasets, more diseases, deep learning, hospital integration, mobile app.

Q47. What are the limitations?

Limited datasets, data quality dependency, not a replacement for doctors.

Most Important Question: Explain your project from start to end.

Problem Statement → Dataset Collection → Preprocessing → EDA → Train-Test Split → Model Training → Evaluation → Save Model → Streamlit Dashboard → Prediction → Conclusion.